AI ENGINEER BOOTCAMP

PRODUCT REVIEW SYSTEM
USING NLP MODELS

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Himma Bootcamps

❯ Start Slide

About The Project

Goal:

Develop an automated product review system using NLP to classify reviews, cluster product categories, and generate recommendations.

Key Tasks:

a. Review Classification
b. Product Category Clustering
c. Review Summarization Using Generative AI

Problem Statement

Manual review analysis is time-consuming and inefficient. We aim to automate this process using NLP models to extract insights and provide product recommendations.

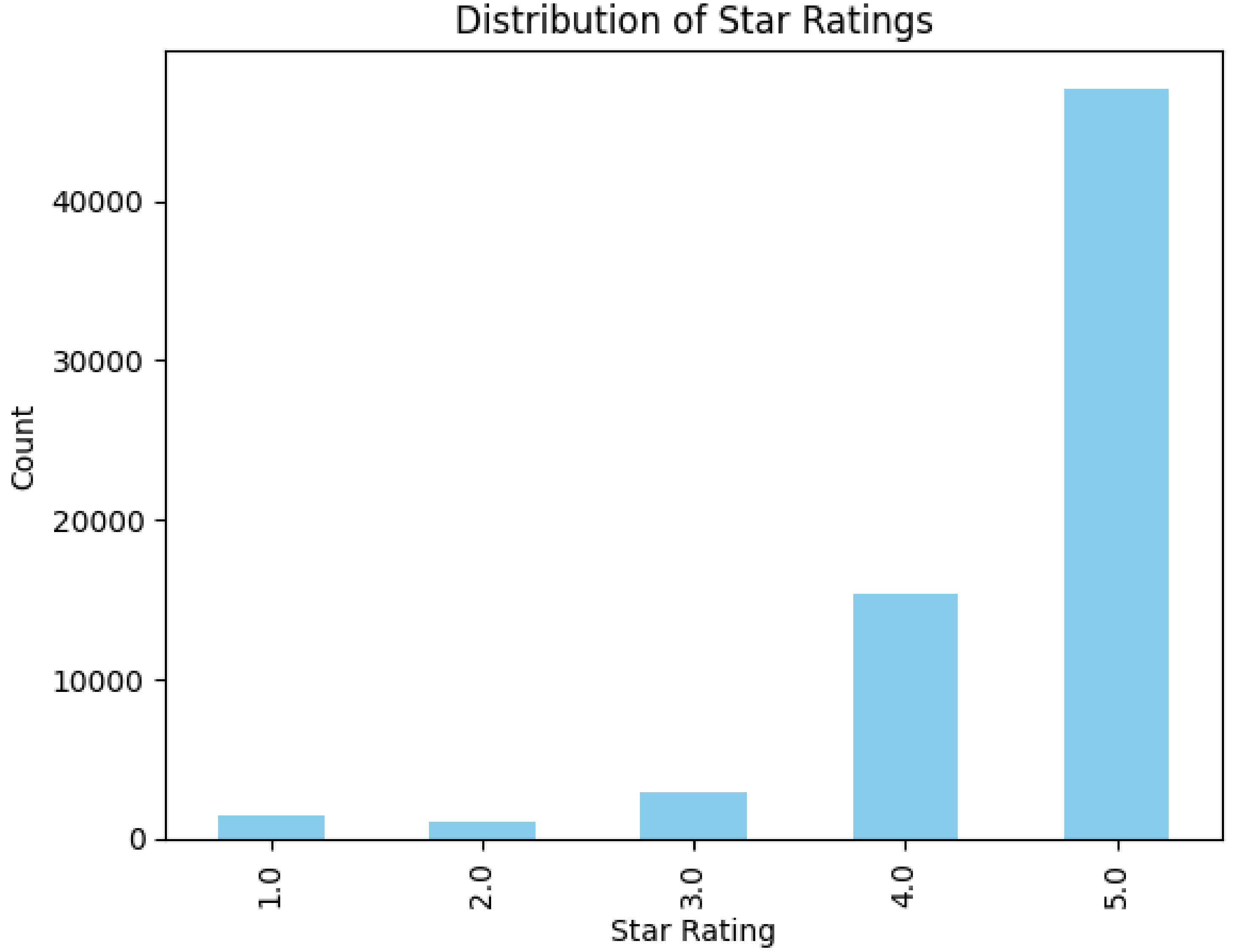
Review Classification

Objective: Classify customer reviews into positive, negative, or neutral to help improve products and services.

Approach:

- Used transformer-based models such as BERT, RoBERTa, and DistilBERT.
- Mapping Star Ratings to Sentiment Classes:
  - 1-2 stars: Negative
  - 3 stars: Neutral
  - 4-5 stars: Positive

- Model Evaluation:
  - Accuracy, Precision, Recall, F1-score
  - Confusion Matrix visualization



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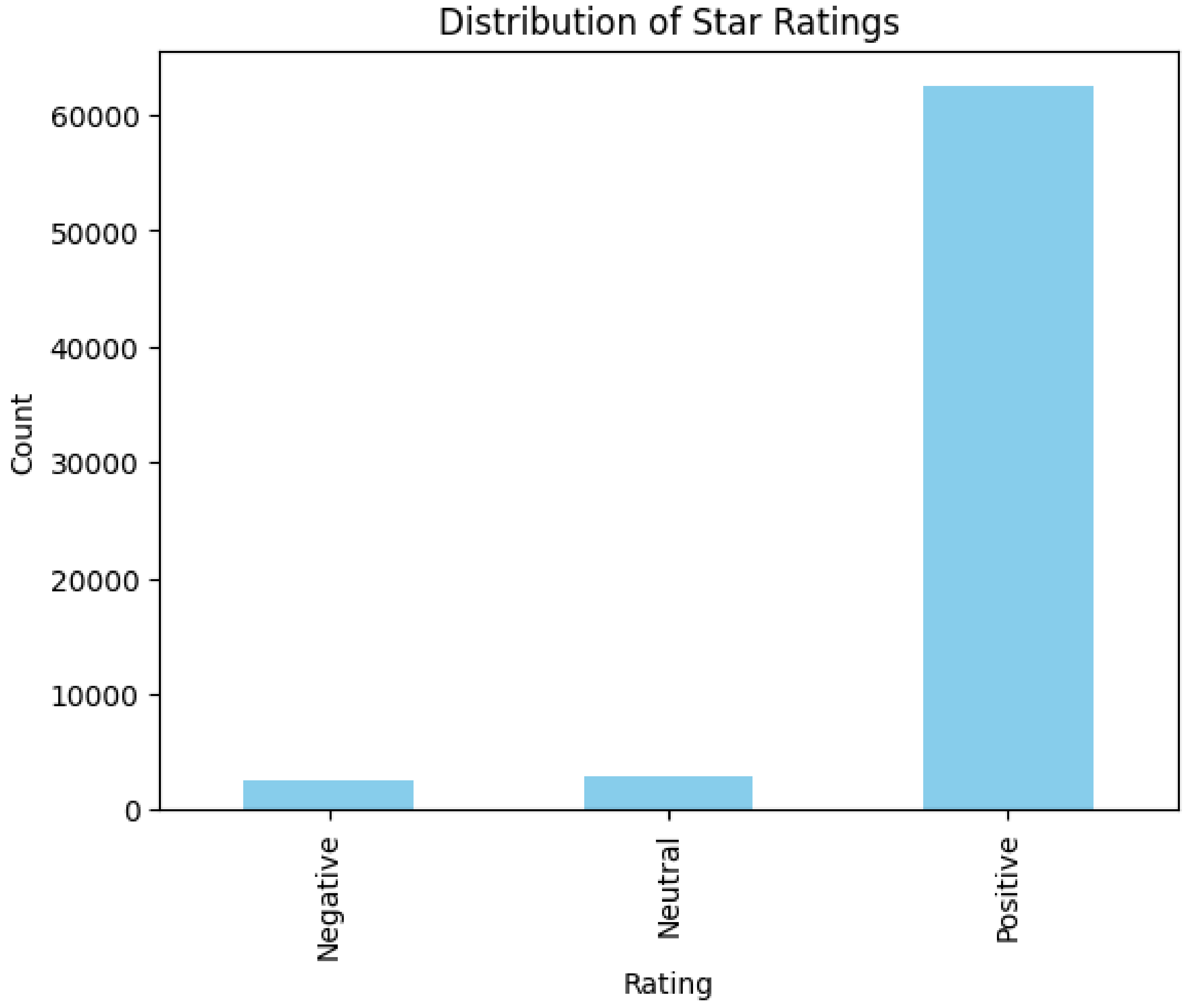

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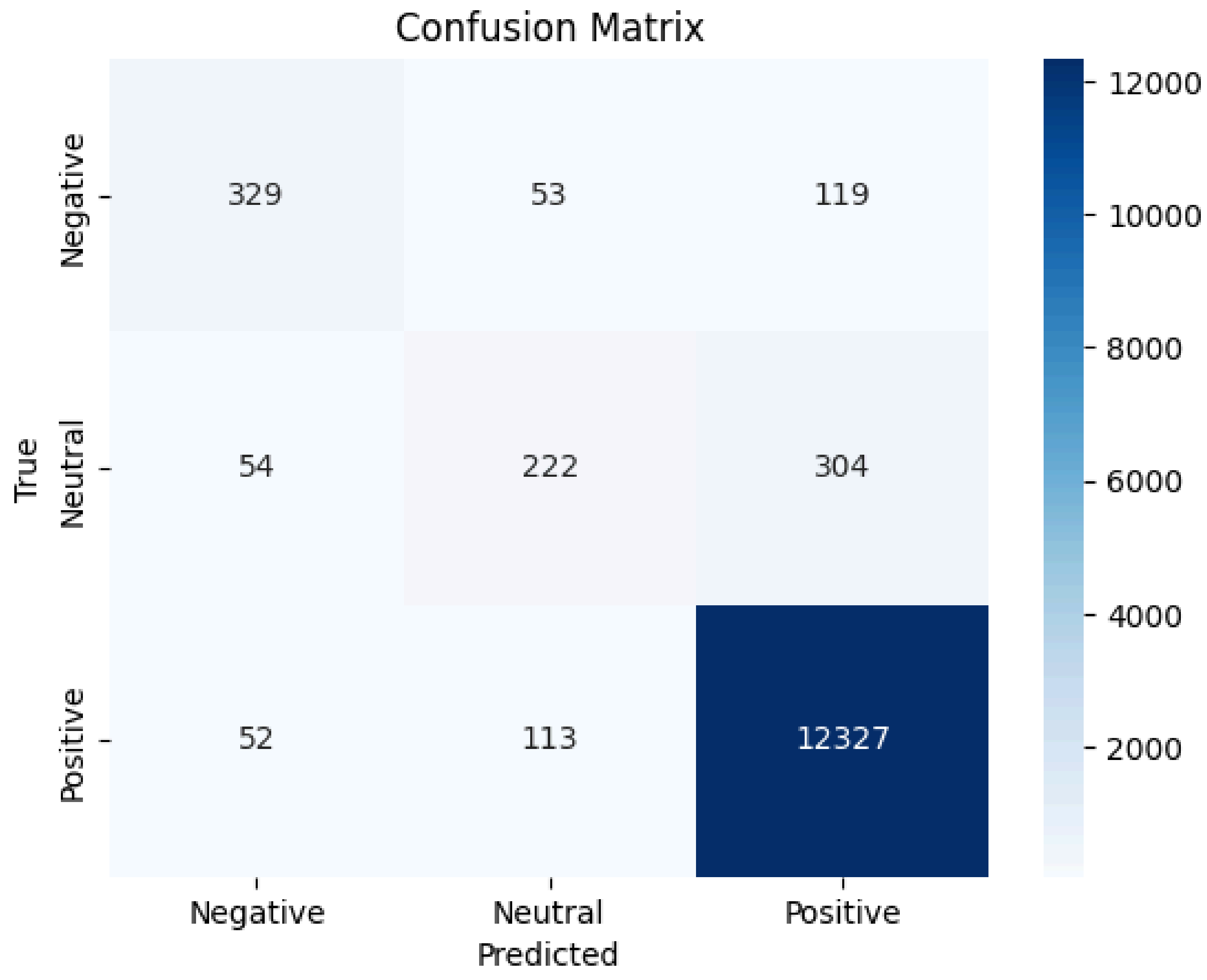

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	0.229900	0.212391	0.934797	0.920445	0.934797	0.921448
2	0.238300	0.202985	0.941354	0.932329	0.941354	0.934900
3	0.178700	0.216550	0.948501	0.939884	0.948501	0.940787
4	0.169000	0.216389	0.948353	0.941593	0.948353	0.943823
5	0.115600	0.227157	0.948795	0.942189	0.948795	0.944466

'bert-base-uncased'

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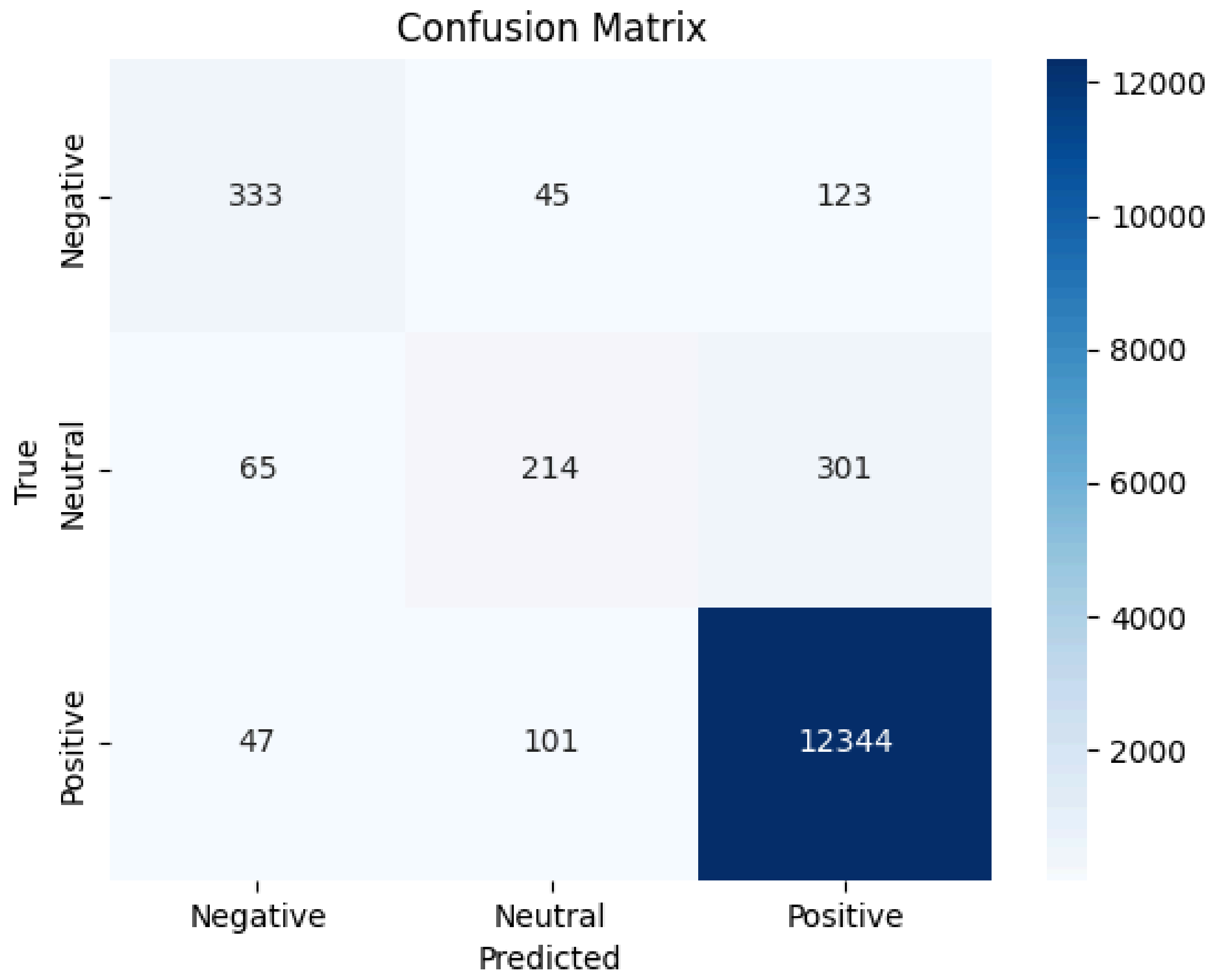

distilbert-base-uncased

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	0.231400	0.210201	0.934944	0.919829	0.934944	0.920202
2	0.231600	0.208206	0.940986	0.931612	0.940986	0.934255
3	0.180400	0.214088	0.947543	0.937821	0.947543	0.939186
4	0.179000	0.219586	0.948574	0.941546	0.948574	0.943416
5	0.121500	0.225156	0.949753	0.942817	0.949753	0.944959

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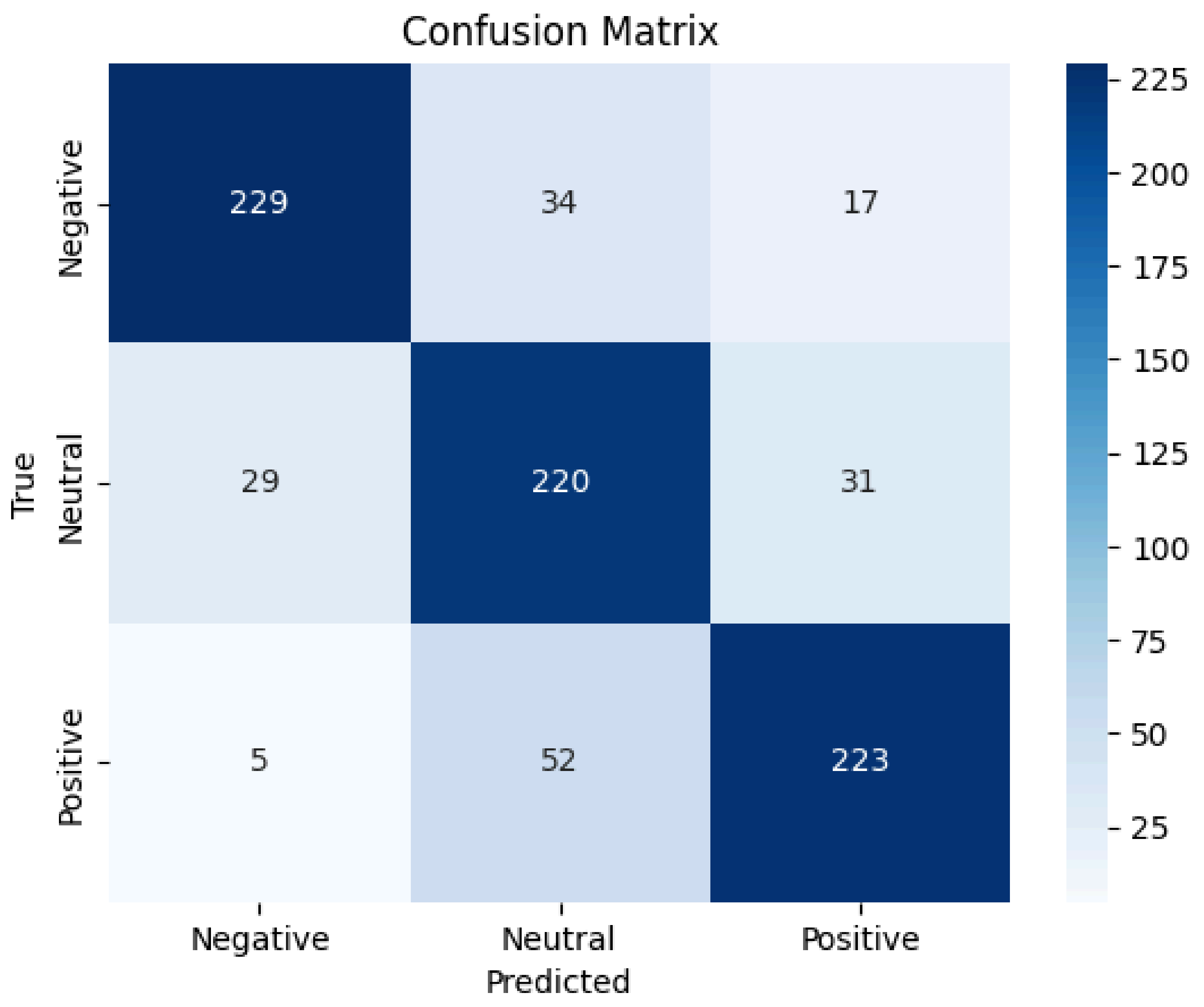

[1050/1050 01:38, Epoch 5/5]

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	0.853100	0.835194	0.610714	0.638039	0.610714	0.605592
2	0.633400	0.707605	0.715476	0.735765	0.715476	0.693237
3	0.457700	0.627854	0.765476	0.778765	0.765476	0.768786
4	0.264000	0.682788	0.780952	0.787164	0.780952	0.782740
5	0.133500	0.798138	0.800000	0.804185	0.800000	0.801251

'bert-base-uncased'

Task1

```
<class 'pandas.core.frame.DataFrame'>
Index: 33237 entries, 0 to 28331
Data columns (total 25 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   id                    33237 non-null  object
 1   dateAdded             33237 non-null  object
 2   dateUpdated           33237 non-null  object
 3   name                  33237 non-null  object
 4   asins                 33237 non-null  object
 5   brand                 33237 non-null  object
 6   categories            33237 non-null  object
 7   primaryCategories     33237 non-null  object
 8   imageURLs             33237 non-null  object
 9   keys                  33237 non-null  object
 10  manufacturer          33237 non-null  object
 11  manufacturerNumber    33237 non-null  object
 12  reviews.date          33237 non-null  object
 13  reviews.dateAdded     1044 non-null   object
 14  reviews.dateSeen      33237 non-null  object
 15  reviews.doRecommend   20991 non-null  object
 16  reviews.id            70 non-null     float64
 17  reviews.numHelpful    21020 non-null  float64
 18  reviews.rating        33237 non-null  int64
 19  reviews.sourceURLs    33237 non-null  object
 20  reviews.text          33237 non-null  object
 21  reviews.title         33224 non-null  object
 22  reviews.username      33231 non-null  object
 23  sourceURLs            33237 non-null  object
 24  reviews.didPurchase   9 non-null      object
dtypes: float64(2), int64(1), object(22)
memory usage: 6.6+ MB
```

Product Category Clustering

Objective: Group product reviews into 4-6 meta-categories to simplify analysis.

Categories Example:

1. Ebook Readers
2. Batteries
3. Accessories (keyboards, stands)
4. Non-electronics (e.g., Nespresso pods)

- Approach: Clustering algorithms like K-Means or Hierarchical Clustering.
- Clustering Results:
- Visualization of clusters (e.g., scatter plot showing different product clusters).

Task2

Use a pre-trained Sentence-BERT model to encode the categories into embeddings

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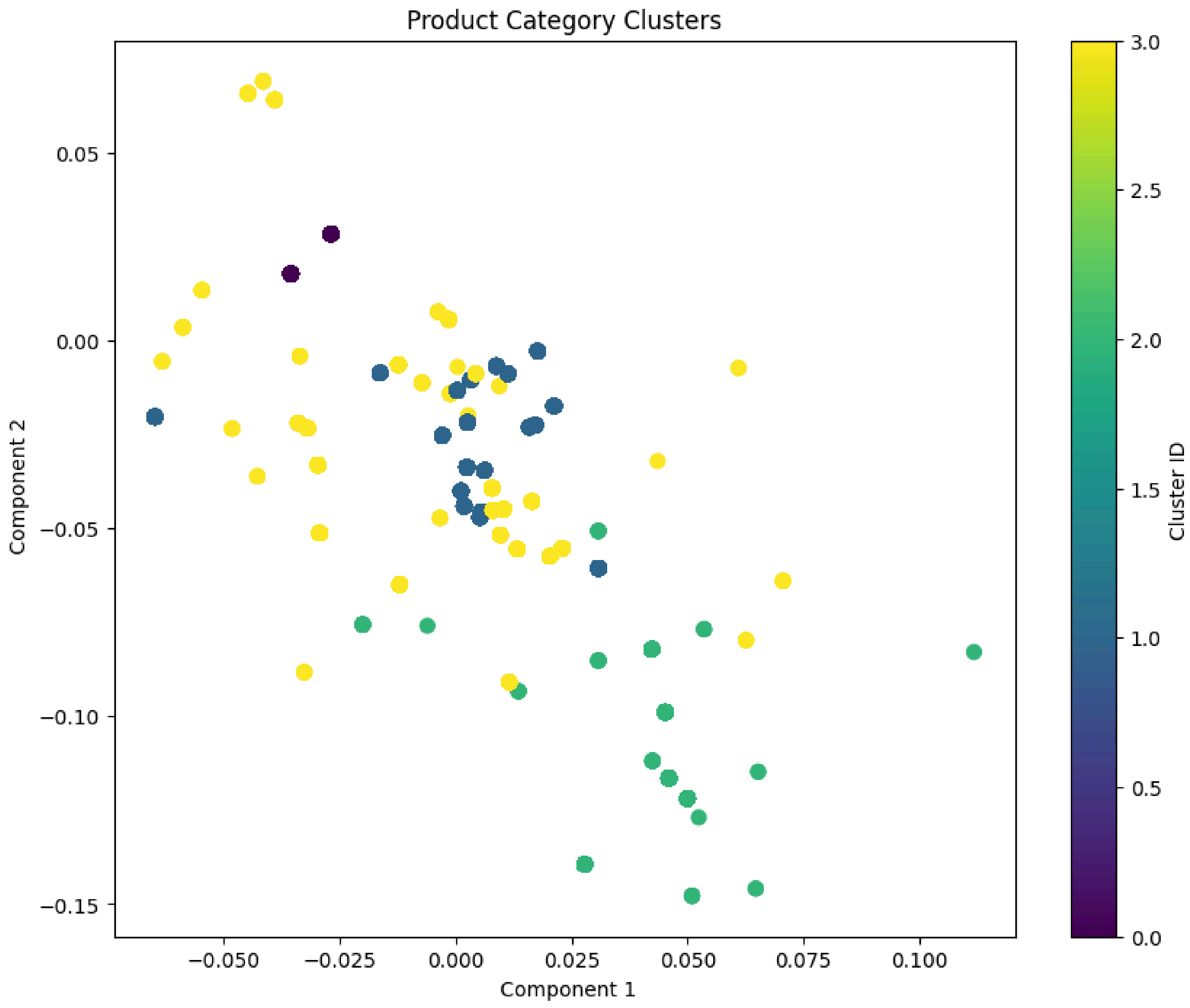


```
cluster_labels = {
    0: "Batteries & Household Electronics",
    1: "Tablets & eBook Readers",
    2: "Smart Home Devices & Audio",
    3: "Tablets & E-readers"
}
```

Task2

Product Category Clustering

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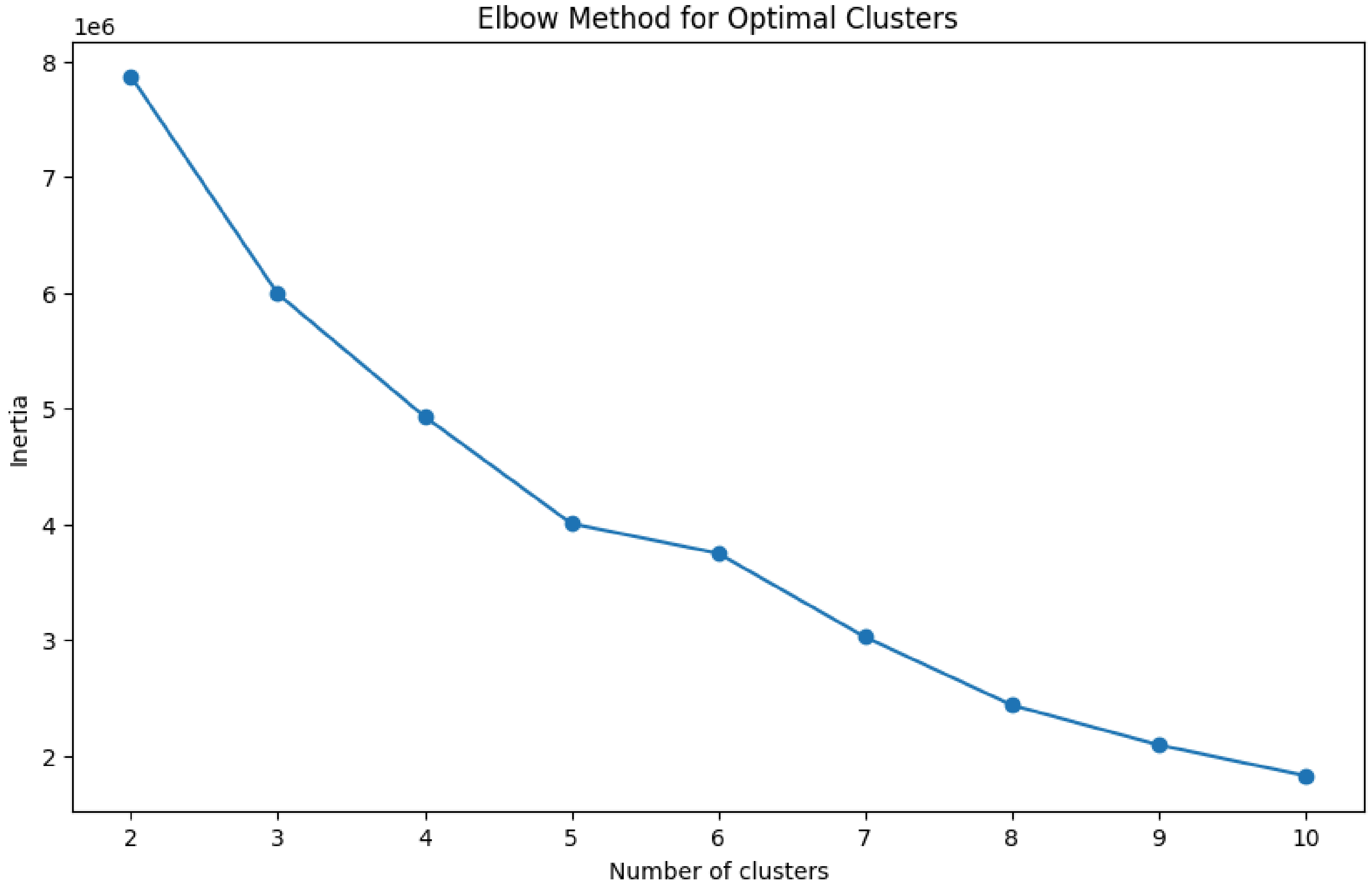

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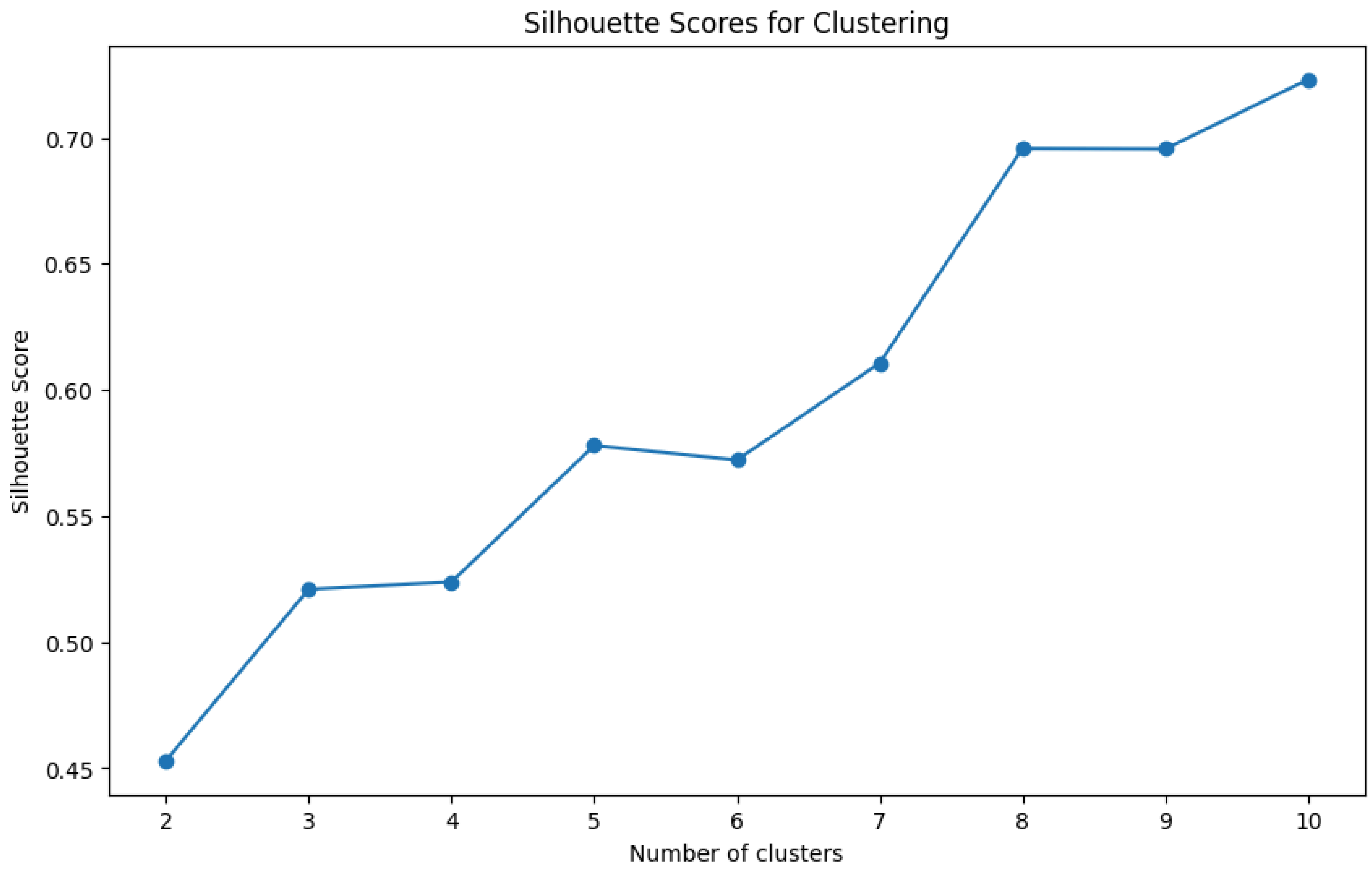

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Product Category Clustering

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Meta-category distribution:

meta_category	count
Tablets & eBook Readers	13368
Batteries & Household Electronics	12071
Tablets & E-readers	5491
Smart Home Devices & Audio	2307

dtype: int64

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Task2

Review Summarization Using Generative AI

- Objective: Summarize reviews into articles that recommend top products for each category.
- Approach: Used models like T5, GPT-3, or BART for generating product summaries.

Summarization Output:

- Top 3 products and key differences
- Top complaints for each product
- Worst product and why to avoid it

Task3

Summarization

```
Summary for Category: Computers,Electronics Features,Tablets,Electronics,iPad & Tablets,Kindle E-readers,iPad Accessories,Used:Tablets,E-Readers

### Top 3 Products in the Category:

1. **Amazon Kindle E-Reader 6" Wifi (8th Generation, 2016)**
   - Key Differences: Lightweight design, easy to use, suitable for outdoor reading.
   - Top Complaints: Lack of color display, limited app functionality, slow page-turning speed.

2. **Apple iPad Pro (12.9-inch, Wi-Fi, 256GB) - Silver (4th Generation)**
   - Key Differences: Large screen size, powerful performance, Apple Pencil compatibility.
   - Top Complaints: Expensive price point, limited file management options, heavy to hold for extended periods.

3. **Samsung Galaxy Tab S7+ 12.4" 256GB WiFi Tablet (Mystic Black) SM-T970NZKEXAR**
   - Key Differences: AMOLED display, S Pen included, DeX mode for desktop-like experience.
   - Top Complaints: Issues with fingerprint sensor accuracy, some software glitches, high price tag.

### Worst Product in the Category:

The worst product in this category is the **Amazon Kindle E-Reader 6" Wifi (8th Generation, 2016)**. Despite its positive reviews for being lightweight and easy to use, it lacks fea

This concise article highlights the top products in the category, their key differences, common complaints, and identifies the worst product to help consumers make informed purchasi

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Summary for Category: Computers,Amazon Echo,Virtual Assistant Speakers,Audio & Video Components,Electronics Features,Computer Accessories,Home & Tools,See more Amazon Echo Show Smar

**Top 3 Products in the Category:**

1. **Amazon Echo Show Alexa-enabled Bluetooth Speaker with 7" Screen**
   - **Key Differences**: Features a 7" screen for visual display, integrated Alexa voice assistant, Bluetooth speaker capabilities.
   - **Top Complaints**: Some users experienced connectivity issues with other smart devices, limited customization options for the screen display.

2. **Amazon Echo Dot (4th Gen) Smart Speaker**
   - **Key Differences**: Compact design, improved sound quality, Alexa voice assistant integration.
   - **Top Complaints**: Sound quality may not be as robust as larger speakers, occasional misinterpretation of voice commands.

3. **Google Home Max Smart Speaker**
```

Task3

Challenges and Solutions

- Challenges:

Unbalanced dataset handling and model optimization.
Balancing accuracy vs. computational efficiency.

- Solutions:

Using pretrained transformer models.
Fine-tuning for better classification and summarization results.

Future Work

Future Work

- Improvements:
- Fine-tuning models for better accuracy.
- Expanding dataset to include reviews from different platforms (e.g., Amazon, Yelp).
- Adding multilingual support to reach a broader audience.
- Additional Features:
- User feedback to continuously improve the model.

Conclusion

- Summary: We built an NLP-powered product review system that classifies, clusters, and summarizes reviews to help businesses gain valuable insights.
- Next Steps: Continue optimizing the models and expand the platform for more product categories and reviews.

THANK YOU